

Final Analysis Insight of Assignment 1

Project Title:

PRODUCT INNOVATION HUB

Introduction: In this report Analyses Product Data set.

Key fields: *Product ID, Product Name, Brand Name, Price, Quantity, Category, Total Price, Price Range, Day, Country Code, Month*

- **Data Summarization:**

SUM, COUNT AND AVERAGE FUNCTIONS

1) SUM FUNCTION: In this function we using calculation of totals, subtotals and summarizing data.

Syntax: **FORMULAE: =SUM (D2:D35*E2:E35)**

Screenshot of Sum Table:



	A	B	C	D	E	F	G	H	I	J	K
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Total Price	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	\$30,000	High Price	28	US	JAN-US
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	\$1,200	High Price	15	US	FEB-US
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	\$5,200	High Price	03	US	MAR-US
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	\$22,500	High Price	11	US	APR-US
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	\$1,400	High Price	22	US	MAY-US
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	\$9,000	High Price	07	UK	JUN-UK
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	\$150	Standard Price	19	UK	JUL-UK
9	23-AUG-UK	Blender	Ninja	\$90	35	Kitchen	\$3,150	High Price	23	UK	AUG-UK
10	05-SEP-UK	Tablet	Apple	\$500	50	Electronics	\$25,000	High Price	05	UK	SEP-UK

Answer: **Total Price of All Products: \$2,89,700**

2) COUNT FUNCTION: Count function is always counting only Numeric value and Integer and Decimal value. The count Function not string text format.

Syntax: **FORMULAE: =COUNTA (B2:B35)**

Screenshot of Count A Table:

N17 =COUNTA(Dataset[Product Name])

	A	B	C	D	E	F	G	H	I	J	K
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Total Price	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	\$30,000	High Price	28	US	JAN-US
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	\$1,200	High Price	15	US	FEB-US
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	\$5,200	High Price	03	US	MAR-US
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	\$22,500	High Price	11	US	APR-US
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	\$1,400	High Price	22	US	MAY-US
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	\$9,000	High Price	07	UK	JUN-UK
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	\$150	Standard Price	19	UK	JUL-UK
9	23-AUG-UK	Blender	Ninja	\$90	35	Kitchen	\$3,150	High Price	23	UK	AUG-UK
10	05-SEP-UK	Tablet	Apple	\$500	50	Electronics	\$25,000	High Price	05	UK	SEP-UK
11	14-OCT-UK	Hiking Boots	Timberland	\$130	10	Outdoor	\$1,300	High Price	14	UK	OCT-UK

Answer: Total Number of Products: 34

- 3) **AVERAGE FUNCTION:** Count function is always counting only Numeric value and Integer and Decimal value. The count Function not string text format.

Syntax: FORMULAE: =AVERAGE (G2:G35)

Screenshot of Average Table:

N32 =AVERAGE(Dataset[Total Price])

	A	B	C	D	E	F	G	H	I	J	K
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Total Price	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	\$30,000	High Price	28	US	JAN-US
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	\$1,200	High Price	15	US	FEB-US
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	\$5,200	High Price	03	US	MAR-US
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	\$22,500	High Price	11	US	APR-US
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	\$1,400	High Price	22	US	MAY-US
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	\$9,000	High Price	07	UK	JUN-UK
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	\$150	Standard Price	19	UK	JUL-UK
9	23-AUG-UK	Blender	Ninja	\$90	35	Kitchen	\$3,150	High Price	23	UK	AUG-UK
10	05-SEP-UK	Tablet	Apple	\$500	50	Electronics	\$25,000	High Price	05	UK	SEP-UK

Answer: Average Price of the Products: \$8,521

- **Data Range Analysis:**

MIN AND MAX FUNCTIONS

- 4) **MIN FUNCTION:** Calculating minimum value of Price Range.

Syntax: FORMULAE: =MIN (G2:G35)

Screenshot of MIN Table:

N45 : X ✓ fx =MIN(Dataset[Total Price])

	A	B	C	D	E	F	G	H	I	J	K
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Total Price	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	\$30,000	High Price	28	US	JAN-US
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	\$1,200	High Price	15	US	FEB-US
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	\$5,200	High Price	03	US	MAR-US
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	\$22,500	High Price	11	US	APR-US
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	\$1,400	High Price	22	US	MAY-US
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	\$9,000	High Price	07	UK	JUN-UK
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	\$150	Standard Price	19	UK	JUL-UK
9	23-AUG-UK	Blender	Ninja	\$90	35	Kitchen	\$3,150	High Price	23	UK	AUG-UK
10	05-SEP-UK	Tablet	Apple	\$500	50	Electronics	\$25,000	High Price	05	UK	SEP-UK

Answer: Minimum Price of All Products: \$150

5) **MAX FUNCTION:** Calculating maximum value of Price Range.

Syntax: FORMULAE: =MAX (G2:G35)

Screenshot of MAX Table:

N53 : X ✓ fx =MAX(Dataset[Total Price])

	A	B	C	D	E	F	G	H	I	J	K
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Total Price	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	\$30,000	High Price	28	US	JAN-US
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	\$1,200	High Price	15	US	FEB-US
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	\$5,200	High Price	03	US	MAR-US
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	\$22,500	High Price	11	US	APR-US
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	\$1,400	High Price	22	US	MAY-US
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	\$9,000	High Price	07	UK	JUN-UK
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	\$150	Standard Price	19	UK	JUL-UK
9	23-AUG-UK	Blender	Ninja	\$90	35	Kitchen	\$3,150	High Price	23	UK	AUG-UK
10	05-SEP-UK	Tablet	Apple	\$500	50	Electronics	\$25,000	High Price	05	UK	SEP-UK

Answer: Maximum Price of All Products: \$36,000

• Logical Function:

IF FUNCTION

6) **IF FUNCTION:** It is fundamental tool. Performing comparisons between test result TRUE or FALSE.

Syntax:

FORMULAE: =IF (G2>=500,"High Price", IF (G2<500,"Standard Price", "Low Price"))

Screenshot of NESTED IF Table:

R9 =IF(G2>=500,"High Price",IF(G2<500,"Standard Price","Low Price"))

	A	B	C	D	E	F	G	H	I	J	K
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Total Price	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	\$30,000	High Price	28	US	JAN-US
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	\$1,200	High Price	15	US	FEB-US
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	\$5,200	High Price	03	US	MAR-US
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	\$22,500	High Price	11	US	APR-US
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	\$1,400	High Price	22	US	MAY-US
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	\$9,000	High Price	07	UK	JUN-UK
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	\$150	Standard Price	19	UK	JUL-UK
9	23-AUG-UK	Blender	Ninja	\$90	35	Kitchen	\$3,150	High Price	23	UK	AUG-UK
10	05-SEP-UK	Tablet	Apple	\$500	50	Electronics	\$25,000	High Price	05	UK	SEP-UK

Answer: Price Range: "High Price"

Syntax: FORMULAE: =IF(G8>500,"High Price", "Standard Price")

Screenshot of IF Table:

R15 =IF(G8>500,"high Price","Standard Price")

	A	B	C	D	E	F	G	H	I	J	K
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Total Price	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	\$30,000	High Price	28	US	JAN-US
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	\$1,200	High Price	15	US	FEB-US
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	\$5,200	High Price	03	US	MAR-US
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	\$22,500	High Price	11	US	APR-US
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	\$1,400	High Price	22	US	MAY-US
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	\$9,000	High Price	07	UK	JUN-UK
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	\$150	Standard Price	19	UK	JUL-UK
9	23-AUG-UK	Blender	Ninja	\$90	35	Kitchen	\$3,150	High Price	23	UK	AUG-UK

Answer: Price Range: "Standard Price"

● Condition Functions:

SUMIF AND COUNTIF FUNCTIONS

7) **SUMIF FUNCTION:** SUMIF allows you to sum value. You need apply range criteria, sum range. Specified range are Summed.

Syntax: FORMULAE: =SUMIF (F2:F35,"Electronics", D2:D35)

Screenshot of SUMIF Table:

R27 : ✕ ✓ *fx* =SUMIF(Dataset[Category],"Electronics",Dataset[Total Price])

	A	B	C	D	E	F	G	H	I	J	K
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Total Price	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	\$30,000	High Price	28	US	JAN-US
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	\$1,200	High Price	15	US	FEB-US
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	\$5,200	High Price	03	US	MAR-US
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	\$22,500	High Price	11	US	APR-US
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	\$1,400	High Price	22	US	MAY-US
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	\$9,000	High Price	07	UK	JUN-UK
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	\$150	Standard Price	19	UK	JUL-UK
9	23-AUG-UK	Blender	Ninja	\$90	35	Kitchen	\$3,150	High Price	23	UK	AUG-UK
10	05-SEP-UK	Tablet	Apple	\$500	50	Electronics	\$25,000	High Price	05	UK	SEP-UK

Answer: Total Electronics category: \$2,34,200

8) **COUNTIF FUNCTION:** Counts the number of “Occurrence” that’s satisfy a given condition. Key point is Conditional counting.

Syntax: **FORMULAE:** =COUNTIF (D2:D35,"<100")

Screenshot of COUNTIF Table:

R35 : ✕ ✓ *fx* =COUNTIF(Dataset[Price (\$)],"<100")

	A	B	C	D	E	F	G	H	I	J	K
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Total Price	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	\$30,000	High Price	28	US	JAN-US
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	\$1,200	High Price	15	US	FEB-US
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	\$5,200	High Price	03	US	MAR-US
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	\$22,500	High Price	11	US	APR-US
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	\$1,400	High Price	22	US	MAY-US
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	\$9,000	High Price	07	UK	JUN-UK
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	\$150	Standard Price	19	UK	JUL-UK
9	23-AUG-UK	Blender	Ninja	\$90	35	Kitchen	\$3,150	High Price	23	UK	AUG-UK
10	05-SEP-UK	Tablet	Apple	\$500	50	Electronics	\$25,000	High Price	05	UK	SEP-UK

Answer: Number of Products with a Price less than: \$11

• Text Functions:

LEFT,RIGHT AND MID FUNCTIONS

9) **LEFT FUNCTION:** Can be used limit the number of characters displayed from left side of text string.

Syntax: **FORMULAE:** =LEFT(A2,2)

Screenshot of LEFT Table

File Home Insert Draw Page Layout Formulas Data Review View Help Acrobat

Paste (Clipboard) Font (Aptos Narrow, 11, Bold, Italic, Underline, Color) Alignment (Left, Center, Right, Merge & Center) Number (General, Percentage, Decimals, Fractions) Condition Formatting

Q47: =LEFT(A2,2)

	A	B	C	D	E	F	G	H	I	J
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	High Price	28	US	JAN-US
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	Standard Price	15	US	FEB-US
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	Standard Price	03	US	MAR-US
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	High Price	11	US	APR-US
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	Standard Price	22	US	MAY-US
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	Standard Price	07	UK	JUN-UK
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	Standard Price	19	UK	JUL-UK
9	23-AUG-UK	Blender	Ninja	\$90	35	Kitchen	Standard Price	23	UK	AUG-UK
10	05-SEP-UK	Tablet	Apple	\$500	50	Electronics	High Price	05	UK	SEP-UK

H
Day
28
15
03
11
22
07
19
23
05

Answer: Day: 28

- 10) **RIGHT FUNCTION:** Can be used the number of characters displayed from right side of text string.

Syntax: FORMULAE: =RIGHT(A2,2)

Screenshot of RIGHT Table

	A	B	C	D	E	F	G	H	I	J
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	High Price	28	US	JAN-US
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	Standard Price	15	US	FEB-US
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	Standard Price	03	US	MAR-US
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	High Price	11	US	APR-US
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	Standard Price	22	US	MAY-US
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	Standard Price	07	UK	JUN-UK
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	Standard Price	19	UK	JUL-UK
9	23-AUG-UK	Blender	Ninja	\$90	35	Kitchen	Standard Price	23	UK	AUG-UK
10	05-SEP-UK	Tablet	Apple	\$500	50	Electronics	High Price	05	UK	SEP-UK

Country Code
US
US
US
US
US
UK
UK
UK
UK

Answer: Country Code: US

11) **MID FUNCTION:** Extract a sub string from middle of text string.

Syntax: FORMULAE: =MID(A2,4,6)

Screenshot of MID Table

The screenshot shows the Microsoft Excel interface. The ribbon is set to 'Home'. The formula bar shows the formula `=MID(A2,4,6)` in cell Q62. Below the ribbon is a table with 10 rows and 10 columns. The columns are labeled: Product ID, Product Name, Brand Name, Price (\$), Quantity, Category, Price Range, Day, Country Code, and Month. The data rows contain product information. Below the main table is a separate list of months: JAN-US, FEB-US, MAR-US, APR-US, MAY-US, JUN-UK, JUL-UK, AUG-UK, and SEP-UK.

	A	B	C	D	E	F	G	H	I	J
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	High Price	28	US	JAN-US
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	Standard Price	15	US	FEB-US
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	Standard Price	03	US	MAR-US
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	High Price	11	US	APR-US
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	Standard Price	22	US	MAY-US
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	Standard Price	07	UK	JUN-UK
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	Standard Price	19	UK	JUL-UK
9	23-AUG-UK	Blender	Ninja	\$90	35	Kitchen	Standard Price	23	UK	AUG-UK
10	05-SEP-UK	Tablet	Apple	\$500	50	Electronics	High Price	05	UK	SEP-UK

Month
JAN-US
FEB-US
MAR-US
APR-US
MAY-US
JUN-UK
JUL-UK
AUG-UK
SEP-UK

Answer: Month: JAN-US

Conclusion: In this Price Analysis the above mentioned using all functions in the excel we will get quick analyses of summary, visualizations of table we can easily understand to analysis in product Innovation Hub.